

Physics-Informed Neural Networks on Bending Moment of Laterally Loaded Piles

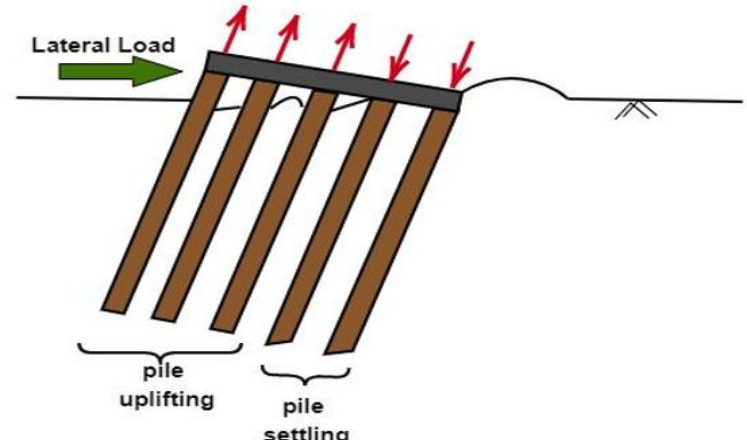
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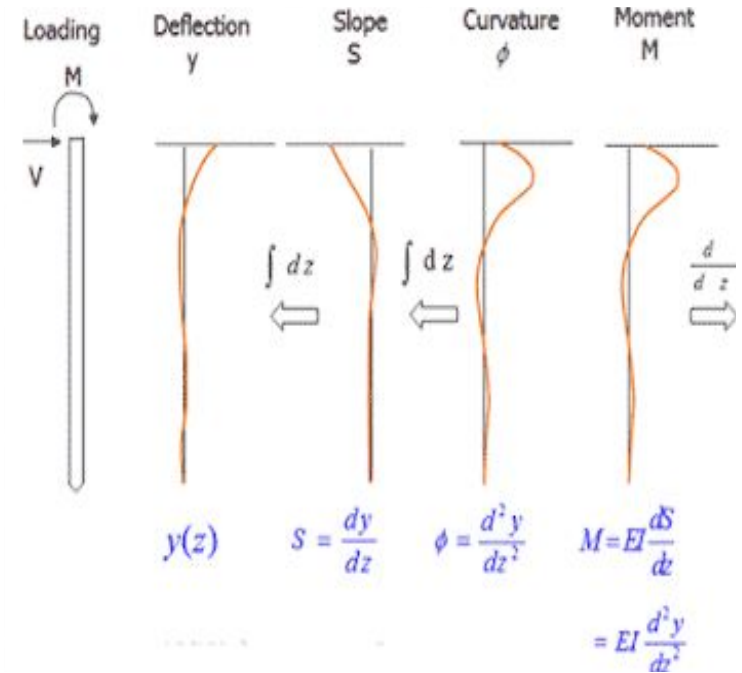


Introduction

- **Bending moments are critical** for assessing the structural integrity of laterally loaded underground structures.
- **Traditional monitoring focuses on displacements**, not internal forces like bending moments.
- **Current methods require complex back-calculation**
 - FEM
 - Continuum models.

Objective

- **PINNs can estimate bending moments**
 - *Very less* Displacement data
 - Boundary conditions.
 - Slope
 - Equations linking deflection and bending moments
- **Assumptions and Considerations**
 - linear elasticity
 - small deformations
 - plane sections remain plane
 - Cantilever beams
 - Perpendicular displacements



Tasks

1. Non-dimensionalize the governing equation

2. Develop Forward PINN model

Pred M given z, EI fixed

$$EI \frac{d^4 y}{dz^4} = p(z) \quad \text{where} \quad p(z) = -ky$$

3. Develop Inverse PINN model

$$EI \frac{d^2 y}{dz^2} = -M(z) \quad \text{and} \quad -p(z) = \frac{d^2 M}{dz^2}$$

Pred M given z, EI as learnable

4. Develop Parametric model (Optional)

Pred M given z, EI

5. Concise presentation

Non-dimensionalization

$$EI \frac{d^2 y}{dz^2} = -M(z)$$

$$\frac{dy}{dz} = \frac{d(y^* \bar{y})}{d(z^* \bar{z})} = \frac{y^*}{z^*} \frac{d(\bar{y})}{d(\bar{z})}$$

$$\bar{z} = \frac{z}{L^*}$$

$$\frac{d^2 y}{dz^2} = \frac{d^2 (y^* \bar{y})}{d(z^* \bar{z})^2} = \frac{y^*}{z^{*2}} \frac{d^2 (\bar{y})}{d(\bar{z})^2}$$

$$\bar{y} = \frac{y}{y^*}$$

$$M^* = EI \frac{y^*}{z^{*2}}$$

$$\bar{M} = \frac{M}{M^*}$$

$$EI \frac{y^*}{z^{*2}} \frac{d^2 y}{dz^2} = -M^* \bar{M} = -\bar{M} \frac{y^*}{z^{*2}}$$

$$\frac{d^2 \bar{y}}{d\bar{z}^2} + \bar{M} = 0$$

Forward PINN Model

Case 1: Reese (1997)

1. Data Loading & Preprocessing

- Read z, y from CSV, sort by depth z
- Non-dimensionalize

2. Boundary Conditions Setup

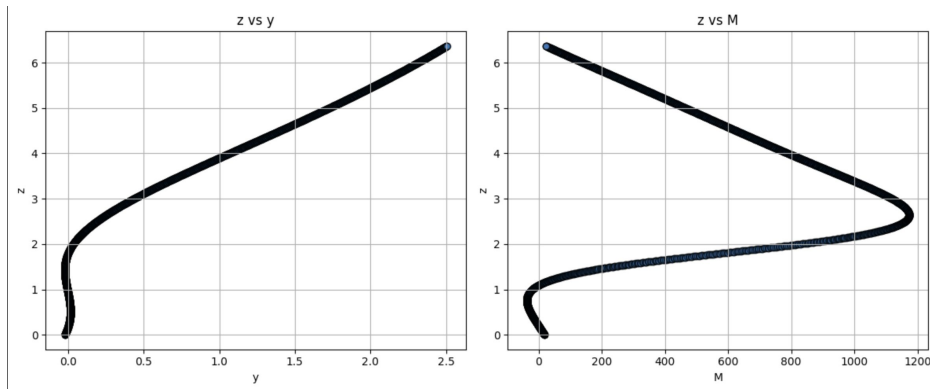
- **Head** : enforce both displacement and slope
- **Toe** : enforce displacement

3. Collocation Points

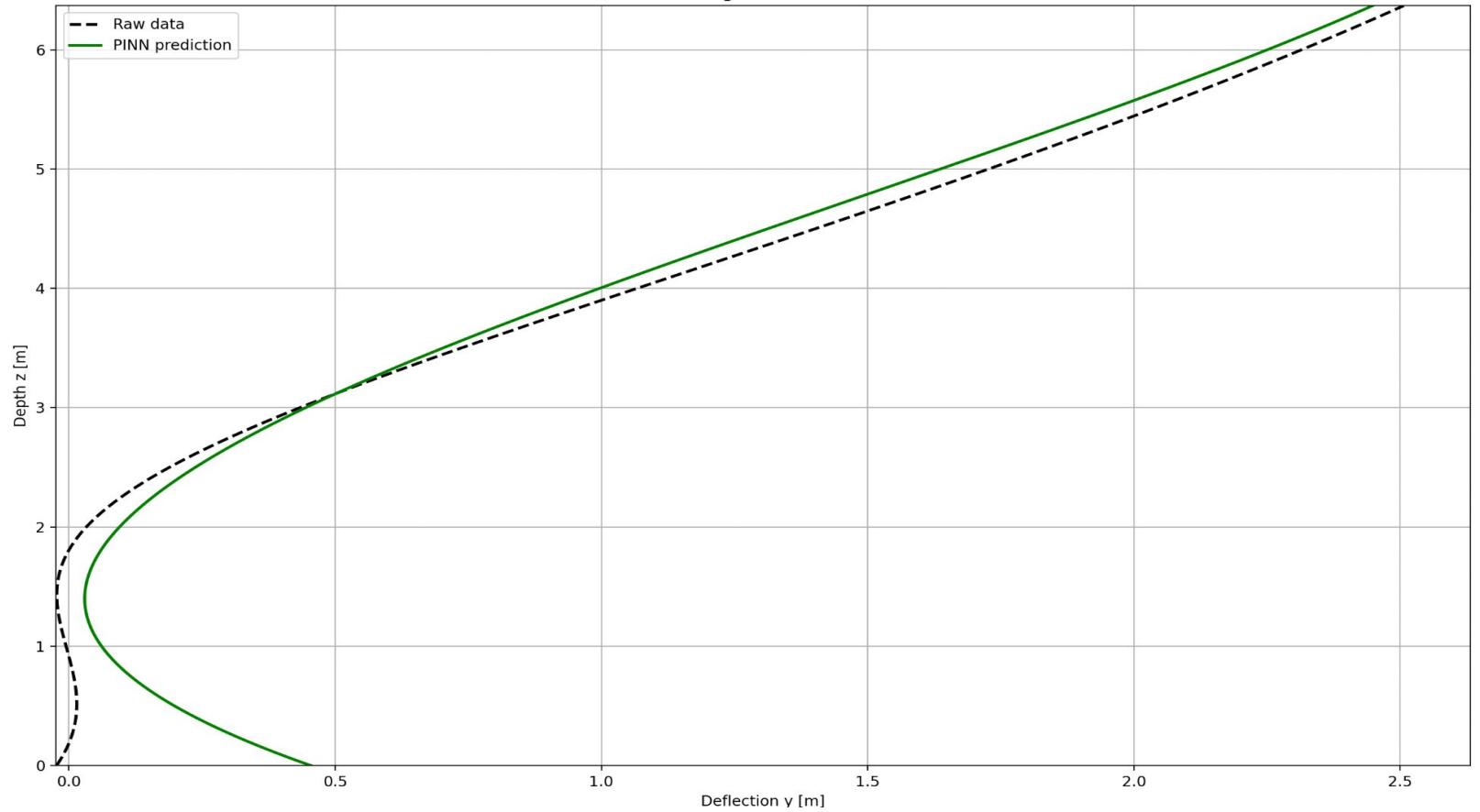
- Uniformly sample = 500 points in the domain $[0,1]$ for evaluating the PDE residual

4. Network Architecture & Parameters

- Fully-connected MLP: 3 hidden layers $\times$ 128 units + Tanh activations



Full-range: Raw vs PINN



5. Physics Residual Definition

- Computes the fourth derivative via nested `-torch.autograd.grad` calls

6. Loss Composition

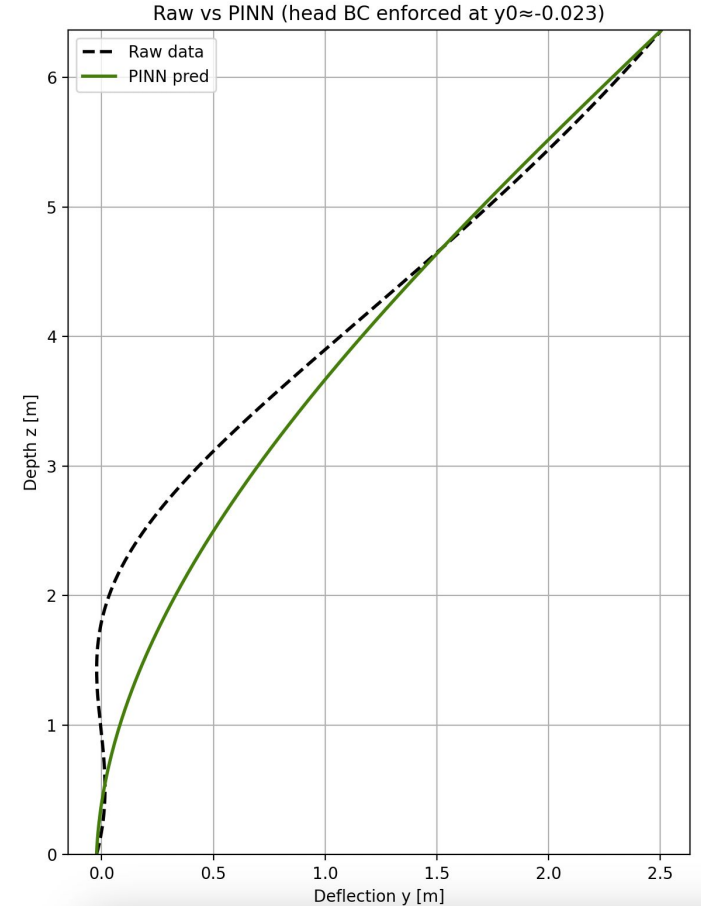
- PDE loss: mean-square of residual at collocation points
- Head BC loss: displacement + slope
- Toe BC loss: displacement

7. Training Strategy

- Stage 1: Adam
- Stage 2: L-BFGS

8. Evaluation & Metrics

- Plot raw vs. PINN-predicted deflection profiles
- Report R2 on the full dataset = 0.95



Objective

Use a Physics-Informed Neural Network to learn the bending moment profile from depth data.

•Data & Pre-processing

- Loaded depth and moment measurements from CSV
- Applied non-dimensionalization for stable training

•Network Architecture

- Fully-connected network with three hidden layers (128 units each) and Tanh activations
- Inputs: normalized depth; outputs: normalized moment

•Physics Integration

- Enforced the governing relationship via automatic differentiation
- No additional data needed, physics guides the fit

•Constraints & Conditions

- Value matching at top and bottom depths
- Slope matching at those boundaries
- Exact match at the midpoint

•Loss Composition

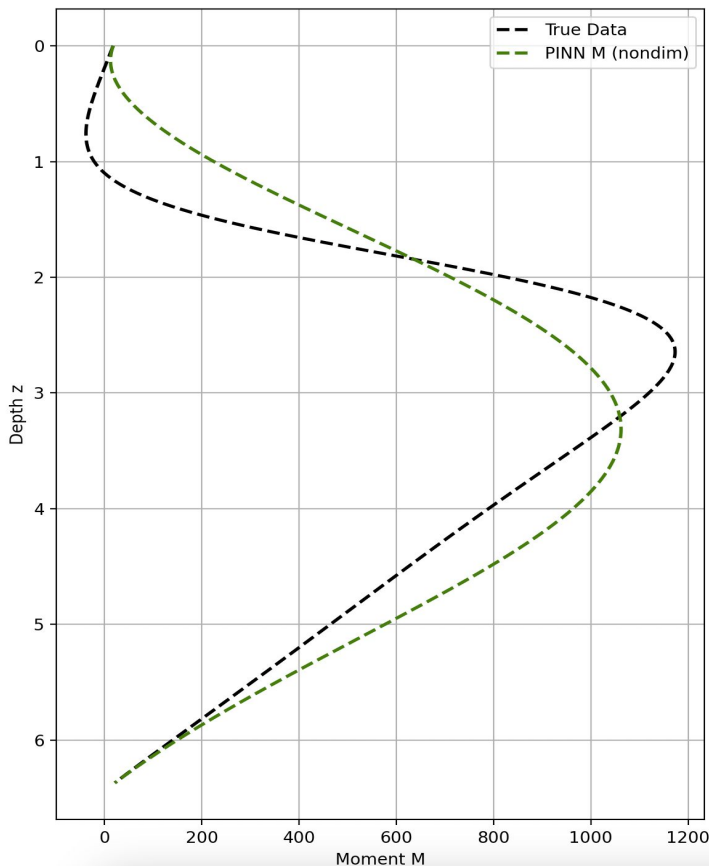
- PDE residual loss + weighted terms for boundary values, slopes, and midpoint

•Training Workflow

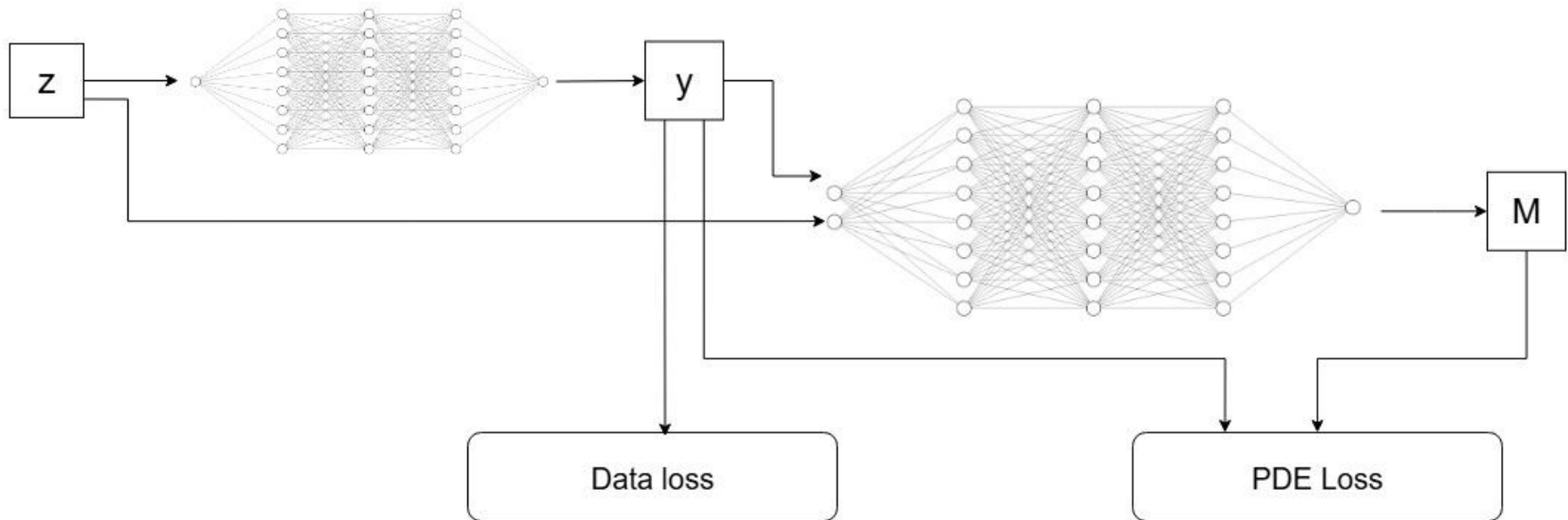
- First: 10 000 epochs of Adam optimizer (learning rate 1e-3)
- Then: L-BFGS quasi-Newton fine-tuning for high precision

•Outcomes

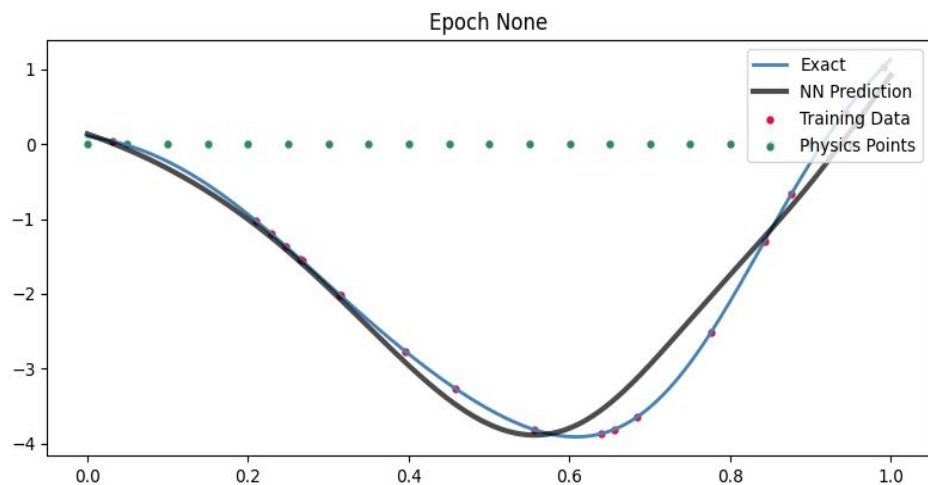
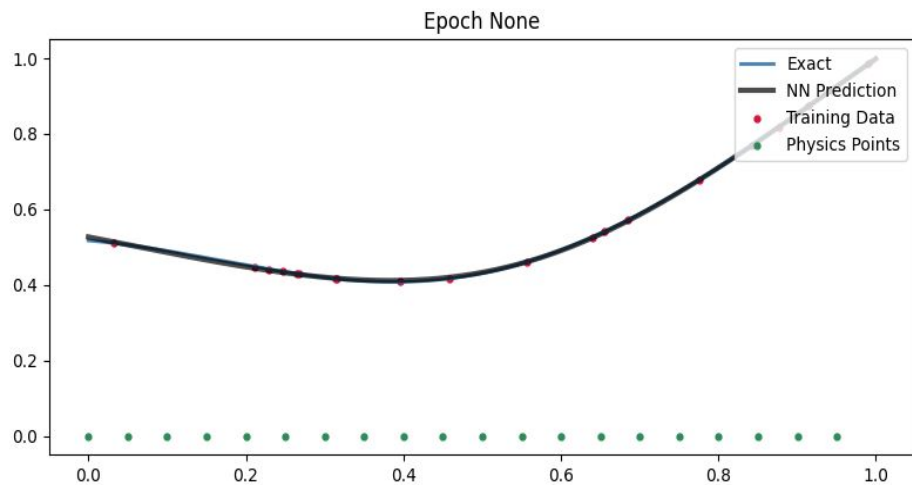
- Plot of true vs. model-predicted moment
- Performance metrics: RMSE and R^2 on original scale



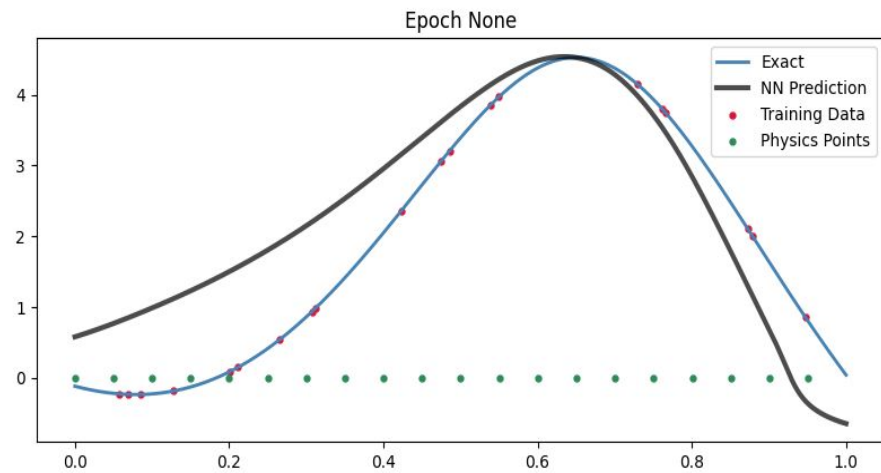
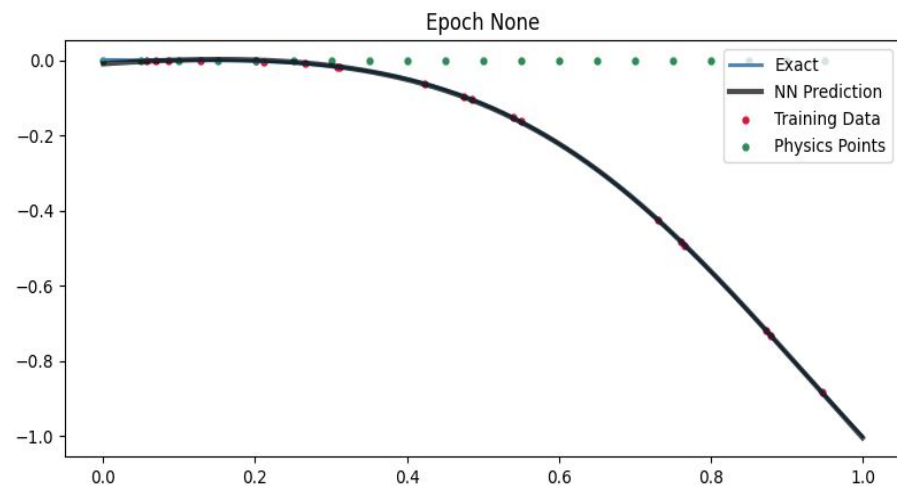
- 20 data points out of 2000 data points.
- No boundary conditions, adaptive learning rate, Tanh as activation function and 10,000 epochs
- 1st NN
 - z coordinates as an input
 - displacement y as output
 - 3 hidden layers, each 32 neurons
- 2nd NN
 - z and disp. y as inputs
 - moment M as output
 - 5 hidden layers, each 32 neurons



Case 2: Cheng et al. (2007)



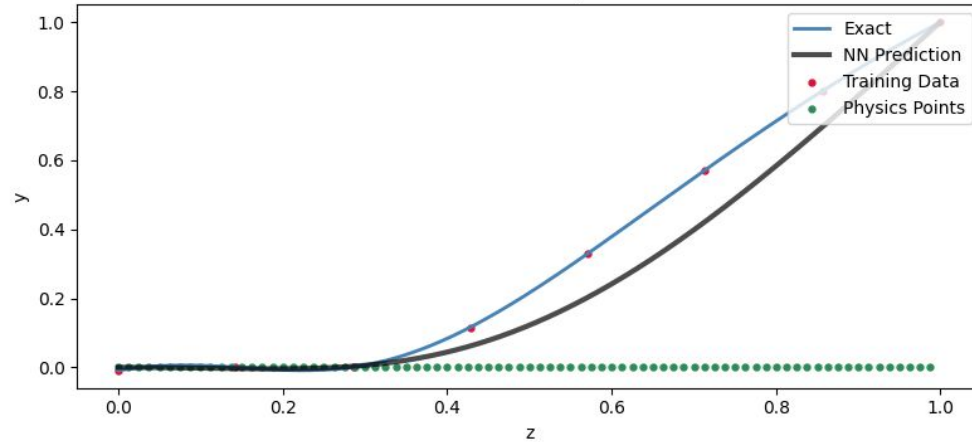
Case 3 : Liyanapathirana and Poulos (2005)



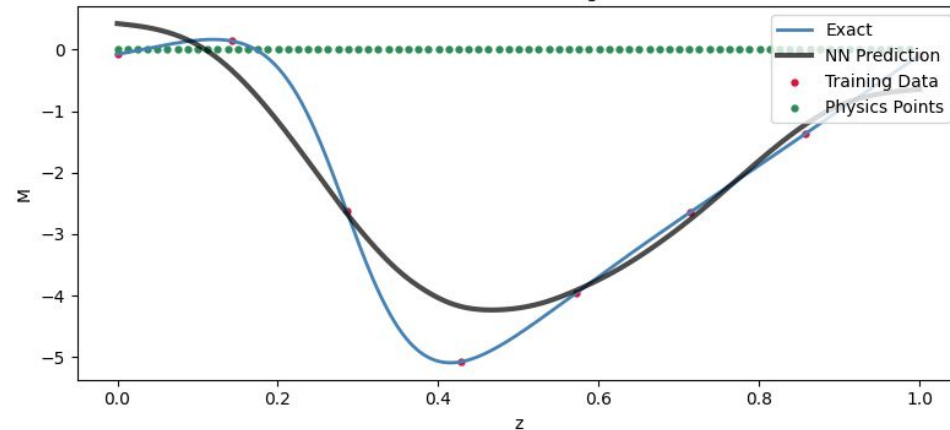
Inverse PINN

- PINN using displacement data, that treats the bending stiffness parameter EI as a learnable quantity within the neural network.
- Hard due to EI not being constant.
- Boundary conditions
- Only 8 measurements of M used
- PDE calculated in 80 points
- True EI $3.7 \cdot 10^6 \text{ N/m}^2$
- Learned EI $1,7 \cdot 10^5 \text{ N/m}^2$

NN Predicted and Exact Displacement Profiles



NN Predicted and Exact Bending Moment Profiles



Conclusion

- PINNs are cool - combining data with physics lets us solve hard problems and model real world scenarios more accurately.
- Even smallest change can break stuff - we learned the hard way that ReLU breaks second derivatives
- Reading units is hard - we lost a whole day chasing a problem that was just a mix-up between mm and meters.
- We gained a deep understanding of how PINNS work, and learned a lot about the loaded piles.